



**End-to-End Air Quality Index (AQI) Prediction  
and  
Real-Time Forecasting System for Lahore**

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**Program:** 10Shine Data Science Internship Program

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**Live Project Link:**

<https://aqi-predictor-u2yvyxwzfzqtskvxkddpkv3.streamlit.app>

**GitHub Repository:**

<https://github.com/hadeebajaved/AQI-Predictor>

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## **1. Abstract**

Air pollution, particularly excessive amounts of particulate matter like PM<sub>2.5</sub> and PM<sub>10</sub>, has been identified as one of the most crucial environmental issues in Lahore. For this project, a full-fledged MLOps pipeline has been developed to predict the value of PM<sub>2.5</sub> in the coming 24 hours and forecast the overall trend in air pollution for the coming 3 days. With a machine learning predictive model using the Random Forest algorithm, MongoDB for storing the features and GitHub Actions for automation, users will gain real-time insights via the interactive Streamlit web app.

## **2. Introduction & Problem Statement**

The worsening condition of air pollution in large cities needs to be tracked via an intelligent monitoring system. While weather apps can show the current condition of air pollution in your city, they fail to provide predictions based on AI and interactions among different pollutants. In this project, a complete self-sufficient machine learning system has been built that not only predicts future PM<sub>2.5</sub> values depending upon PM<sub>10</sub>, CO, NO<sub>2</sub>, and time-based features but also does the job of retraining itself every day.

### 3. Architecture and Technology

To ensure high scalability and minimize manual intervention, this system transitions from a conventional static model into a dynamic, automated framework. This evolution is supported by the following core components:

**Data storage (Backend):** A NoSQL architecture via MongoDB is employed to manage and store all historical atmospheric datasets efficiently.

**Continuous Integration/Continuous Deployment:** GitHub Actions are utilized to orchestrate two primary automated workflows:

**Hourly Pipeline:** Retrieves live sensor data to update the database on an hourly basis.

**Daily Pipeline:** Facilitates the regular retraining of the AI model with fresh data every 24 hours.

**Machine Learning:** Predictive modeling is powered by the Scikit-Learn library, specifically utilizing the Random Forest Regressor.

**Front-end Application:** An interactive user interface is deployed through the Streamlit Cloud platform.

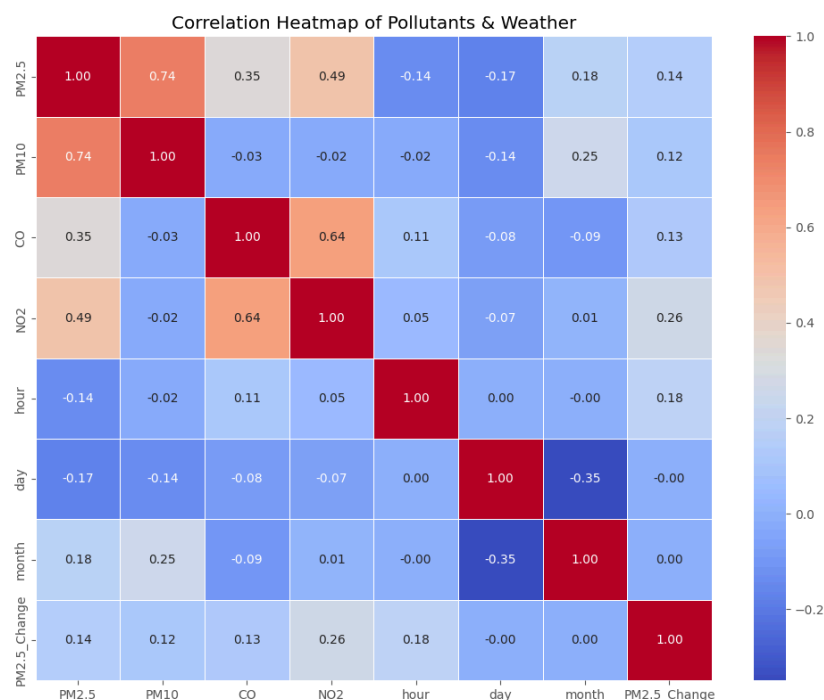
|                                                                                                                                                                                                |                                                                                          |                                                                                                                                                                                                  |     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
|  <b>Hourly Feature Pipeline</b><br>Hourly Feature Pipeline #4: Manually run by <a href="#">hadeebajaved</a> |  main |  1 minute ago<br> 45s  | ... |
|  <b>Daily Model Training</b><br>Daily Model Training #3: Manually run by <a href="#">hadeebajaved</a>       |  main |  2 minutes ago<br> 39s | ... |

## 4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted before model training in order to identify the hidden trends in the atmospheric data.

### 4.1. Correlation Analysis

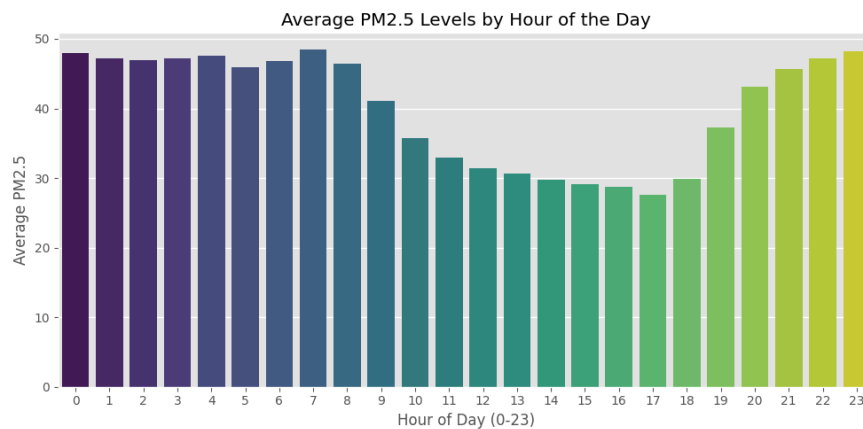
From the correlation heatmap analysis, it was identified that there was a very high positive correlation between PM10 and PM2.5 concentration. It was also noticed that NO2 concentration also had a substantial effect on particulate matter due to vehicle emissions.



### 4.2. Time-Series Analysis & Hourly Trend

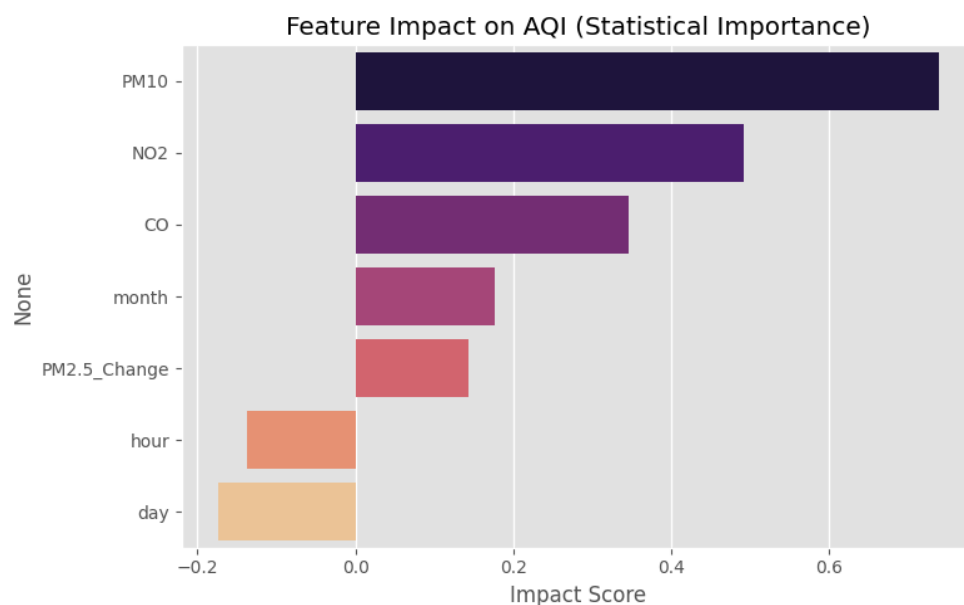
On conducting time-series analysis for 24 hours, different trends were discovered. PM2.5 concentration was observed to have sharp peaks

during late night and early morning hours due to temperature inversion effect and heavy traffic in Lahore.



### 4.3. Feature Importance

Feature importance was calculated using Random Forest model which indicated that historical PM2.5 concentration, current PM10 concentration, and particular hour of the day (considering seasonality and daily routine activities) were important features.



## 5. Machine Learning Approach

**Algorithm Selection:** Random Forest Regressor was selected based on its good performance in predicting non-linear data and dealing with complicated relationships among multiple pollutants without overfitting.

**Feature Generation:** Time-related features like hours, days, and months were generated. Moreover, PM2.5\_Change was generated as a feature for monitoring the rate of pollution build-up.

**Evaluation:** Evaluation is an ongoing process performed during the pipeline's execution every day. RMSE and MAE are observed before registering and deploying the new model (model.pkl).

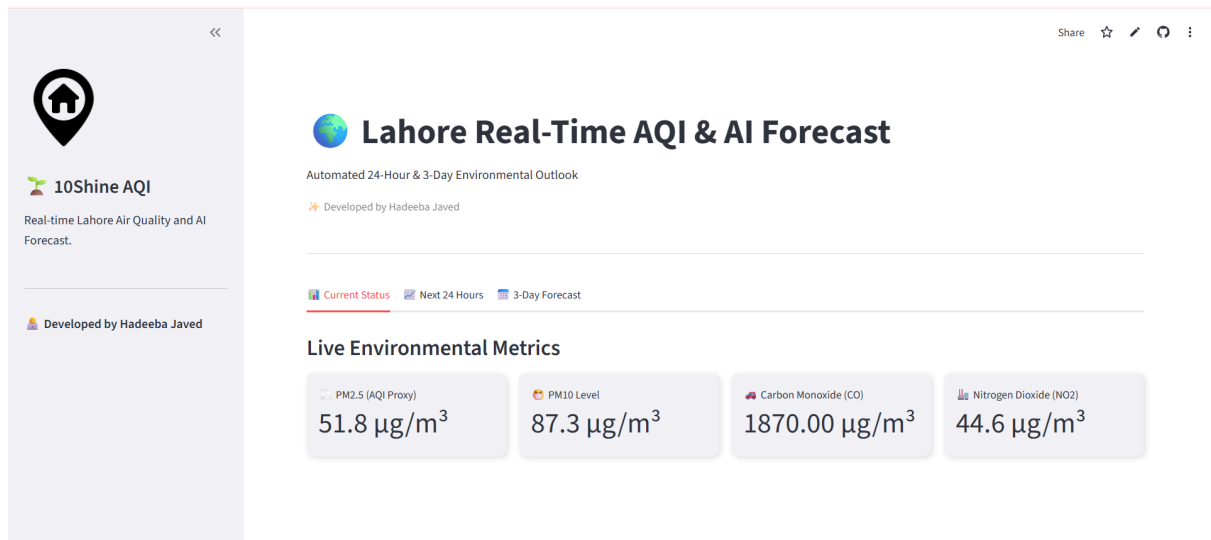
## 6. Web Dashboard Functionality

This frontend app converts complicated machine learning predictions into useful data for all audiences. The web dashboard is composed of the following three modules:

**Live Environmental Measurements:** Shows real-time values of PM2.5, PM10, CO, and NO2 measurements presented in metrics format.

**Next 24 Hours Projection:** This Plotly interactive area chart represents the AI-projected hourly trend, which allows making plans for safe outdoor activities.

**3-Day Projection:** The prediction model represented through color code (Good – Green; Moderate – Yellow; Unhealthy – Red).



## 7. Conclusion

This project successfully demonstrates the full lifecycle of a Data Science solution—from data extraction and modeling to cloud deployment and CI/CD automation. By integrating predictive AI with automated data pipelines, the system provides a highly scalable and maintenance-free solution for real-time environmental monitoring in Lahore.